Steven Thao

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Education

McGill University
GPA: 3.96/4.0

BEng, Computer Engineering

- **McGill In-Course Scholarship**: Awarded to students in the **top 10%** of the engineering faculty. exp. grad: Dec 2027

Experience

Full-Stack Developer Intern
Sept 2025 - Present

Stingray Digital, *Chatter team*

- Building a **full-stack** customer engagement platform for **5,000+ business locations**, enabling owner–client communication, analytics, and AI-assisted review management.
- Developed modern UI features in **React and TypeScript**, improving responsiveness and usability.
- Refactored backend services in **Python/Django REST** Framework with **PostgreSQL**, cutting redundant API calls and improving **query performance** by **up to 50%** through **Redis**-based caching, batching, and schema tuning.
- Researched and optimized **AI retrieval pipelines** using **pgvector** and **HNSW** indexing, running performance tests and analyzing embedding information variance to guide design decisions presented to the product CEO.
- Integrated a **chat-bot** via **Retrieval-Augmented Generation** (RAG) and OpenAI's **AgentKit**, enabling context-aware client interactions with scalable vector search.
- Enhanced reliability through **unit and mini-tests** for backend endpoints, light database maintenance, and **bash scripting** to automate QA validation workflows.
- Supported **AWS-based CI/CD environments**, ensuring smooth feature rollouts, stable **deployment pipelines**, and consistent performance across **distributed services**.

Project Lead
Nov 2024 - Present

McGill AI Lab

- **Led a team of 12 students** in designing and implementing ML models (**logistic/softmax regression, CNNs, SVMs, random forests, autoencoders**) in **NumPy**, achieving classification **accuracies of up to 98%** on benchmark datasets.
- **Authored** a comprehensive **50+ page LaTeX document** detailing **mathematical foundations** and implementation strategies, serving as a key reference for students.
- **Optimized ML models** on the **MNIST dataset** using **cross-validation** and **L1/L2 regularization**, improving classification **accuracy by 8%** and reducing **overfitting**.

Radio systems team member
Sept 2025 – Present

McGill Rocket Team

- Configured and implemented **MQTT communication** between the **ground station** and **Teensy 4.1 microcontroller** via **Mosquitto broker**, enabling telemetry data transfer.
- Assisted in testing and debugging **radio communication protocols**, improving **link reliability and latency performance**.

Projects

Semester project - ECSE321 Course

- Developed a **full-stack** board game management application with **Java** for the backend and **TypeScript** with **Svelte** for the frontend.
- Implemented **RESTful** APIs to manage game data and user interactions.
- Added an **RDBMS** with **Spring Boot** and **JDBC** to interact with a **SQL** database.
- Adopted **Agile** methodologies including **Scrum meetings** to plan, develop, and refine project features.
- Employed **JUnit** for comprehensive unit testing, enhancing code reliability and performance.

Neural Network - MNIST Classifier
GitHub Repository

- Developed a fully connected **neural network (MLP)** from scratch using **Python, NumPy**, **and PyTorch**, achieving **95% accuracy** on MNIST.
- **Enhanced** model **performance** through **hyperparameter tuning, dropout regularization**, and **learning rate optimization**.
- Built real-time **visualizations** with **Matplotlib**, analyzing gradient updates, loss curves, and classification results.

Semester Project - ECSE223 Course
GitHub Repository

- Developed a **full-stack Java application** to assist school administrators in managing students' school supplies.
- Implemented the **MVC architecture** with **UML-based** design, ensuring modularity and scalability.
- Wrote extensive **unit tests** using **JUnit** and **Cucumber/Gherkin**, ensuring code reliability and **+90%** test coverage.
- Designed a responsive UI with **FXML** and **JavaFX**, providing an intuitive user experience.

AutoBot AI Chatbot
Devpost Project Page

Winner of McGill CodeJam 14, Nov. 2024

- Developed an **AI-powered chatbot** that assists car dealerships in recommending vehicles to customers based on conversational data.
- Engineered a **custom recommendation algorithm**, integrating **named entity recognition (NER)** and **sentiment analysis**.
- **Optimized** NLP models using **word embeddings** and **aspect-based sentiment analysis**, improving response accuracy.
- **Won 1st place** in the Matador AI-sponsored challenge, outperforming 10+ teams.

Skills

Programming: Bash, C, Python, JavaScript, SQL, VHDL, Java, ARM Assembly, Typescript, Svelte, React
Hardware: Arduino, FPGA, Raspberry Pi, ESP32
Tools: Altium Designer, LTSpice, Intel Quartus, PyTorch, NumPy, Git
Software: JetBrains Suite, ModelSim, VSCode, Logisim